1 Introduction

1.1 Background and Context

Public transportation systems constitute a critical pillar of modern urban infrastructure. Among all transit modes, buses serve the largest share of daily commuters worldwide, particularly in rapidly urbanising developing nations [1]. Accurate information regarding passenger occupancy is important for improving route planning, optimizing fleet management, monitoring overcrowding, and enhancing passenger safety.

Traditional APC systems commonly employ hardware based sensors such as infrared beams, pressure mats, or ticketing systems. Although these approaches are widely used, they often suffer from several limitations, including high installation and maintenance costs, sensitivity to environmental conditions, and reduced accuracy when multiple passengers enter or exit simultaneously [2].

Recent advancements in computer vision and deep learning have enabled the development of camera based passenger counting systems. Since many buses are already equipped with surveillance cameras for security purposes, vision based methods offer a cost effective and scalable solution [3]. Deep learning models encompassing both object detection and density estimation approaches have achieved state of the art performance on crowd counting benchmarks, demonstrating that neural networks can reliably estimate occupancy in complex visual scenes.

However, bus interiors present unique challenges that distinguish them from general crowd scenes. Passengers frequently occlude each other, surveillance cameras are typically mounted on the ceiling causing perspective distortion, and passenger heads often appear as small objects in the captured images.

1.2 Research Problem

Despite significant progress in crowd counting and object detection, accurately estimating the number of passengers inside buses remains a challenging task. Existing crowd counting approaches, such as CSRNet[4] and DM-Count [5] are primarily designed for outdoor scenes and benchmark datasets such as ShanghaiTech[6], UCF-QNRF[7], and NWPU-Crowd[8]. Although these methods can estimate the total number of people in an image, they do not explicitly identify individual passengers and cannot maintain temporal consistency across video frames.

On the other hand, object detection based methods can detect individual passengers but often struggle in bus environments because of severe occlusion and the small size of passenger heads. Furthermore, frame by frame detection may produce unstable counts and double counting errors when the same passenger is detected repeatedly across consecutive frames.

Although object tracking algorithms such as DeepSORT[9] and ByteTrack[10] have shown promising results in general multi object tracking tasks, their effectiveness in improving passenger counting stability and reducing duplicate counts in bus interiors remains insufficiently explored. Therefore, there is a need to investigate whether head detection can accurately count passengers, determine which detection model performs best for small head detection, and evaluate the contribution of object tracking to stable passenger counting.

1.3 Research Objectives

The main objective of this study is to develop and evaluate a vision based passenger counting system for bus interiors using head detection and object tracking.

The specific objectives are as follows:

- To investigate whether head detection can be used to count passengers accurately in bus interiors.
- To compare different object detection models and determine which model performs best for detecting small passenger heads in real bus scenes.
- To evaluate the effectiveness of object tracking techniques in improving counting stability and reducing double counting across video frames.

1.4 Scope and Limitations

1.4.1 Scope

This study focuses on the development and evaluation of a vision based passenger counting system for bus interiors using deep learning based head detection and object tracking techniques. The proposed system utilizes surveillance video captured from cameras installed inside buses to estimate the number of passengers in real time.

The scope of this study includes:

- Developing a passenger counting framework based on head detection rather than full body detection to better handle crowded bus environments and severe occlusion.
- Evaluating and comparing multiple object detection models for detecting small passenger heads in bus surveillance videos.
- Investigating the effectiveness of multi object tracking algorithms in maintaining passenger identities across consecutive video frames and reducing double counting errors.

This study is limited to passenger counting inside buses and does not consider other public transportation systems such as trains, subways, or airports.

1.4.2 Limitations

Although the proposed approach aims to provide an accurate and robust passenger counting system, several limitations exist.

- The system relies on video captured from ceiling mounted surveillance cameras. Different camera positions or viewpoints may affect detection and tracking performance.
- Severe occlusion remains a challenging problem. When passengers are completely hidden behind others, head detection may fail, leading to undercounting.

- The performance of the tracking algorithm depends heavily on the quality of the object detector. Missed detections or false detections may cause identity switches or fragmented trajectories.
- Variations in lighting conditions, shadows, motion blur caused by bus movement, and image quality may negatively affect the detection accuracy.
- The proposed system assumes that the surveillance camera remains fixed. Camera movement or vibration beyond normal bus operation is not considered in this study.
- The study focuses on passenger counting only and does not address related tasks such as passenger identification, behavior analysis, age estimation, or demographic analysis.

Despite these limitations, the study is expected to provide valuable insights into the feasibility of using head detection and object tracking for accurate passenger counting in bus environments and serve as a foundation for future research in intelligent transportation systems.

1.5 Significance of the Study

This study is significant for both academic research and practical applications. From an academic perspective, the research contributes to the growing field of vision based passenger counting by investigating the integration of head detection and object tracking in bus environments. Unlike conventional crowd counting methods that rely on density estimation, this study focuses on detecting and tracking individual passengers, thereby addressing issues such as occlusion, small object detection, and counting stability.

From a practical perspective, an accurate passenger counting system can provide valuable information for intelligent transportation systems. Real time passenger occupancy data can assist transport authorities in optimizing bus schedules, reducing overcrowding, improving service quality, and enhancing passenger safety. Furthermore, because many buses are already equipped with surveillance cameras, the proposed approach can potentially be deployed without requiring additional hardware, making it a cost effective solution for public transportation systems.

The findings of this study may also contribute to the development of smart city applications where real-time monitoring and data-driven decision-making are essential.

2 Research Motivation

The rapid growth of urban populations has increased the demand for efficient and intelligent public transportation systems. Bus services remain the most accessible and widely used transit option, particularly in developing economies where formal rail infrastructure is limited or nascent. Accurate information about passenger occupancy is essential for improving service quality, optimizing fleet management, reducing overcrowding, and enhancing passenger safety.

Traditional automatic passenger counting systems rely on dedicated hardware sensors, such as infrared beams or pressure sensitive devices. Although these systems are widely used, they often require expensive installation and maintenance and may produce inaccurate counts in crowded situations. Camera based alternatives avoid these pitfalls by exploiting surveillance infrastructure that most modern buses already carry, requiring only software-side upgrades.

Recent progress in deep learning-based object detection and crowd counting has significantly improved counting performance in complex environments. Nevertheless, bus interiors present challenges that are not adequately addressed by existing methods. Passengers are frequently occluded, heads occupy only a small number of pixels, and camera viewpoints differ substantially from those in commonly used crowd counting datasets. Moreover, frame-by-frame detection approaches may produce unstable results due to missed detections or repeated counting of the same passenger.

These challenges motivate the development of a vision-based passenger counting system that combines head detection and object tracking. Head detection focuses on the most consistently visible part of the human body in crowded environments, while tracking can maintain passenger identities across consecutive frames, reducing count fluctuations and duplicate counts.

3 Problem Statement

Accurate passenger counting is a fundamental requirement for intelligent public transportation systems. Reliable passenger statistics enable transport authorities and bus operators to optimize routes, monitor occupancy levels, improve passenger safety, and make data-driven operational decisions. Conventional passenger counting systems, such as infrared sensors, pressure sensors, and ticketing-based methods, often suffer from inaccuracies due to sensor malfunction, passenger overlap, and simultaneous boarding or alighting events.

Recent advances in computer vision and deep learning have demonstrated promising results for crowd counting and object detection in surveillance applications. However, applying these methods to bus passenger counting remains challenging because of the unique characteristics of bus interiors. Passengers are often densely packed, causing severe occlusion where individuals partially or completely block one another. In addition, surveillance cameras are usually installed on the ceiling, resulting in perspective distortion and making passenger heads appear small, especially in the rear sections of the bus. Variations in illumination, motion blur caused by vehicle movement, and frequent passenger movement further complicate accurate counting.

Most existing crowd counting methods have been developed and evaluated on public datasets such as ShanghaiTech[6], UCF-QNRF[7], and NWPU-Crowd[8], which mainly consist of outdoor scenes or general crowd environments. These datasets do not adequately represent the visual characteristics and challenges of bus interiors. Furthermore, many studies focus on estimating crowd density rather than identifying individual passengers across video frames, which may lead to unstable counts and double-counting problems in video-based applications.

Therefore, there is a need to investigate whether head detection can provide accurate passenger counts in bus environments, determine which detection models are most effective for detecting small passenger heads, and examine whether integrating object tracking can improve counting stability and reduce duplicate counts across frames.

4 Research Question

- Can head detection be used to count passengers accurately in bus interiors?

- Which detection model performs best for small-head detection in real bus scenes?
- Can tracking improve counting stability across frames and reduce double counting?

5 Contribution

- Comparison of multiple state of the art object detection architectures for small head detection in real bus interior.
- Design a privacy-preserving head counting system with anonymity at the feature extraction level (not post processing).
- Rigorous analysis of head counting accuracy loss under varying crowd densities (sparse, moderate, crowded) in the bus environment.
- Integrate multi object tracking methods to address the temporal consistency and double-counting problems.

6 Literature

Automatic Passenger Counting systems are essential for public transport systems as they help track passenger movement, increase efficiency and inform decision making. These technologies are extensively deployed in buses, trains and metro services for real-time occupancy estimation. Historically, passenger counts have been recorded manually or from ticketing systems, which can be inaccurate, labour intensive and not real-time. As a consequence, there has been a growing interest in research on automatic passenger counting systems.

6.1 Crowd Counting Using Deep Learning

One of the earliest and most influential works is the Multi-Column Convolutional Neural Network (MCNN)[11], which introduced the ShanghaiTech[6] dataset and employed multiple convolutional columns to handle variations in crowd scale. MCNN demonstrated that deep neural networks could effectively estimate crowd density in highly congested scenes.

Subsequent studies sought to improve counting accuracy by introducing more sophisticated architectures. CSRNet replaced multi-column networks with dilated convolutions to enlarge the receptive field while preserving spatial resolution, achieving state of the art performance on several benchmark datasets[4] . Later approaches, such as DM-Count and Transformer-based crowd counting models, further improved performance by incorporating advanced loss functions, attention mechanisms, and long-range feature modeling [5, 12].

Although these methods achieve excellent results on public crowd counting datasets, their performance in bus environments remains uncertain because bus interiors exhibit unique challenges, including severe occlusion, restricted viewpoints, and small object sizes.

6.2 Vision Based Passenger Counting

Vision based passenger counting methods can generally be categorized into detection-based and density estimation-based approaches. Detection-based approaches identify individual passengers or body parts using object detectors and estimate the passenger count by counting detected instances. Early methods employed handcrafted features combined with classical machine learning algorithms. However, recent developments in deep learning have significantly improved detection accuracy through Convolutional Neural Networks (CNNs) and Transformer-based architectures [13].

Density estimation-based approaches, on the other hand, generate a density map representing the spatial distribution of people in an image. The total count is obtained by integrating the density map over the entire image. These methods are particularly effective in highly crowded scenes where individual detection is difficult [11].

Despite their success, density estimation methods do not explicitly identify individuals and are therefore less suitable for video-based applications where maintaining temporal consistency is important.

6.3 Object Tracking for Stable Passenger Counting

Object tracking aims to maintain the identity of detected objects across consecutive video frames. In passenger counting applications, tracking can prevent double counting and reduce fluctuations caused by missed detections [14].

Recent multi object tracking algorithms, such as DeepSORT[9] and ByteTrack[10], combine motion information with appearance features to associate detections over time. DeepSORT extends the SORT algorithm by incorporating deep appearance descriptors, whereas ByteTrack improves tracking robustness by utilizing both high-confidence and low-confidence detections.

Integrating object tracking with head detection has the potential to improve passenger counting accuracy and stability in dynamic bus environments. However, relatively few studies have systematically investigated the effectiveness of tracking for bus passenger counting, particularly under conditions of severe occlusion and small-head detection.

6.4 Research Gap

The literature reveals several research gaps in vision-based passenger counting. First, most crowd counting studies focus on outdoor crowd datasets such as ShanghaiTech[6], UCF-QNRFQNRF[7], and NWPU-Crowd[8], which do not adequately represent bus interior environments. Second, although head detection has shown promise in crowded scenes, there is limited research comparing different detection architectures for small-head detection in buses. Third, the impact of integrating object tracking with head detection for stable passenger counting remains insufficiently explored.

Therefore, this research aims to address these gaps by investigating the effectiveness of head detection for passenger counting in bus interiors, comparing different detection models for small head detection, and evaluating whether object tracking can improve counting stability and reduce double counting across video frames.

References

[1] International Transport Forum, *ITF Transport Outlook 2023*. Paris: OECD Publishing, 2023. [Online]. Available: https://doi.org

[2] C. McCarthy, H. Ghaderi, F. Martí, P. Jayaraman, and H. Dia, "Video-based automatic people counting for public transport: On-bus versus off-bus deployment," *Computers in Industry*, vol. 164, p. 104195, 2025. [Online]. Available: https://doi.org/10.1016/j.compind.2024.104195

[3] C. McCarthy, H. Ghaderi, P. P. Jayaraman, and H. Dia, "A deep learning approach to real-time video analytics for people and passenger counting," in *Handbook on Artificial Intelligence and Transport*, ser. Research Handbooks in Transport Studies series, H. Dia, Ed. Edward Elgar Publishing, 2023, ch. 12, pp. 348–379. [Online]. Available: https://doi.org/10.4337/9781803929545.00022

[4] Y. Li, X. Zhang, and D. Chen, "Csrnet: Dilated convolutional neural networks for understanding the highly congested scenes," 2018. [Online]. Available: https://arxiv.org/abs/1802.10062

[5] B. Wang, H. Liu, D. Samaras, and M. Hoai, "Distribution matching for crowd counting," 2020. [Online]. Available: https://arxiv.org/abs/2009.13077

[6] L. Zhu, C. Li, Y. Zhongguo, K. Yuan, and S. Wang, "Crowd density estimation based on classification activation map and patch density level," *Neural Computing and Applications*, vol. 32, 05 2020.

[7] H. Idrees, M. Tayyab, K. Athrey, D. Zhang, S. Al-Maadeed, N. Rajpoot, and M. Shah, "Composition loss for counting, density map estimation and localization in dense crowds," *arXiv preprint arXiv:1808.01050*, 2018.

[8] Q. Wang, J. Gao, W. Lin, and X. Li, "Nwpu-crowd: A large-scale benchmark for crowd counting and localization," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 6, pp. 2141–2149, 2020.

[9] A. Pujara and M. Bhamare, "Deepsort: Real time multi-object detection and tracking with yolo and tensorflow," in *2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS)*, 2022, pp. 456–460.

[10] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "Bytetrack: Multi-object tracking by associating every detection box," 2022. [Online]. Available: https://arxiv.org/abs/2110.06864

[11] Y. Zhang, D. Zhou, S. Chen, S. Gao, and Y. Ma, "Single-image crowd counting via multi-column convolutional neural network," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2016, p. 589–597. [Online]. Available: http://dx.doi.org/10.1109/CVPR.2016.70

[12] Z. Cui, Y. Li, and L.-N. Tran, "A transformer-based multimodal fusion model for efficient crowd counting using visual and wireless signals," 2025. [Online]. Available: https://arxiv.org/abs/2504.20178

[13] Y.-W. Hsu, Y.-W. Chen, and J.-W. Perng, "Estimation of the number of passengers in a bus using deep learning," *Sensors*, vol. 20, no. 8, p. 2178, Apr. 2020. [Online]. Available: http://dx.doi.org/10.3390/s20082178

[14] L. V. Ma, M. I. Hussain, J. Park, J. Kim, and M. Jeon, "Adaptive confidence threshold for bytetrack in multi-object tracking," in *2023 12th International Conference on Control, Automation and Information Sciences (ICCAIS)*. IEEE, Nov. 2023, p. 370–374. [Online]. Available: http://dx.doi.org/10.1109/ICCAIS59597.2023.10382403